VaaniQ

Cross-Lingual, Compression-Robust Detection and Calibrated Reliability Estimation for AI-Generated Voice in Indian Languages

Master Presentation Document
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Build: 3c82462d84bf

1. Executive Summary

VaaniQ is a research-grade capstone system that detects AI-generated (cloned and TTS) voice in Hindi, Marathi, and Tamil. Unlike simple deepfake demos, VaaniQ studies WhatsApp-style Opus simulation, calibrated confidence, explainability, and cross-lingual transfer under a reproducible machine-evaluation protocol.

This document is the master reference for capstone presentation, viva voce, and publication submission preparation. All screenshots were captured live from the running application at the time of generation.

Component	Status
Software stack (API + UI + live + explain)	Complete and operational
Training corpus	2346 evaluation instances across Hindi, Marathi, Tamil (4.1 hours)
Corpus provenance	speaker-disjoint Kathbath + IndicSynth publication subset
Speaker-disjoint split	Verified
Held-out test accuracy	91.61%
Held-out test EER	6.56%
Detection on real publication clip	REAL (99% confidence)
Detection on fake publication clip	FAKE (96% confidence)
Held-out test ECE	0.026
GPU acceleration	None

Approved Round 3 Canonical Results

All headline metrics below are copied from artifacts/final_results_manifest.json (commit 084bd47ca6ca1b69a7cdbf424e2946f3794c2a95). Acoustic Baseline V1 is an acoustic embedding plus AASIST-compatible head, not canonical AASIST and not XLS-R. Frozen XLS-R improved ranking performance while classification performance remained broadly comparable.

Experiment	Status	Canonical result
Baseline V1	COMPLETE	n=584; acc 91.61%; P 85.48%; R 98.11%; F1 91.36%; EER 6.56%; AUC 0.9729; min-DCF 0.7841; threshold 0.5
Frozen XLS-R main	COMPLETE	facebook/wav2vec2-xls-r-300m frozen mean-pool; n=584; acc 92.12%; EER 6.88%; AUC 0.9828; min-DCF 0.3144
RQ1 acoustic	COMPLETE	Clean 93.84% / Opus 16 kbps WhatsApp-style simulation 89.38% (n=292)
RQ1 frozen XLS-R	COMPLETE	Clean 91.44% / Opus 92.81% (n=292); codec impact is model-dependent
RQ2 English-only	COMPLETE	54.8% acc; 76.56% EER; 0.162 AUC; all REAL at threshold 0.5
RQ3 LOO	COMPLETE	Hindi 78.83% / 21.83% EER; Marathi 93.29% / 7.14%; Tamil 93.94% / 6.35%

RQ4 calibration	COMPLETE	Val-selected per-language-and-condition; test ECE 0.0245 → 0.026
Benchmark V2	PARTIAL	External-source pilot; source probe 98.48%; does not solve source confounding
FLEURS unseen-real	PILOT	n=9; 55.6% retained only as pipeline validation
Generator-disjoint	PENDING	n=0; no result claimed
Faithful RawNet2	PENDING	Approximate baseline exists; faithful implementation remaining

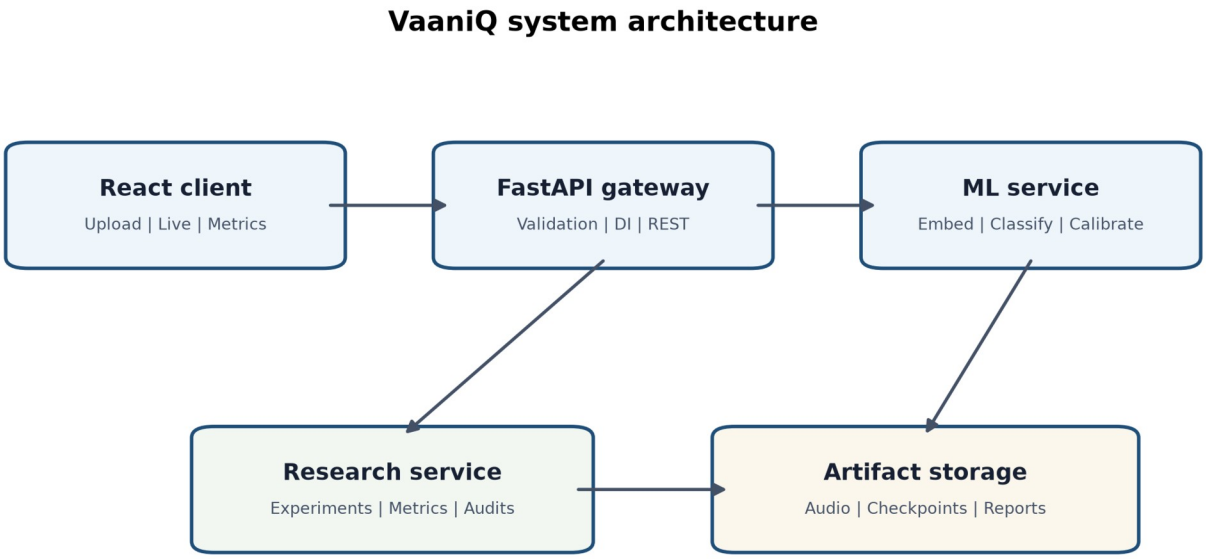
2. Research Questions and Objectives

- RQ1: How much does WhatsApp-style Opus compression degrade multilingual deepfake detectors versus clean audio?
- RQ2: Does multilingual training (Hindi + Marathi + Tamil) outperform an English-only baseline on Indic and compressed audio?
- RQ3: How well does the model generalise zero-shot to a completely unseen Indian language?
- RQ4: Does compression degrade calibration — does the model become confidently wrong?

Objectives map to dataset construction, compression simulation, model benchmarking, cross-lingual evaluation, calibration, live demonstration, and reproducible publication.

3. System Architecture

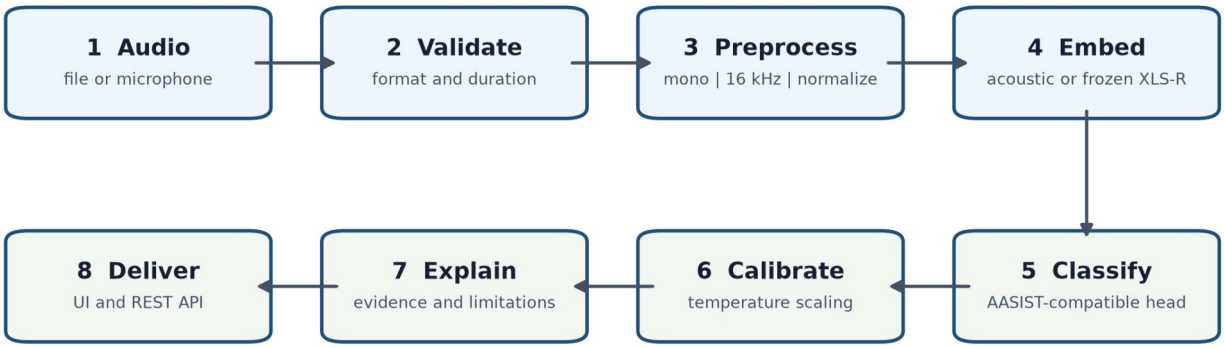
VaaniQ follows clean hexagonal architecture. The React web application communicates with a FastAPI backend. Domain logic is isolated from framework code. Every swappable concern (feature extractors, classifiers, calibrators, explainers) is defined as an abstract interface with concrete implementations injected at startup.



End-to-end ML pipeline:

1. Audio ingestion (upload, file drop, or live microphone PCM16 stream)
2. Validation (MIME type, magic bytes, duration, file size limits)
3. Preprocessing (16 kHz mono, peak normalisation, silence handling)
4. Optional Opus compression twin for robustness evaluation
5. Acoustic embedding extraction (demo path) or frozen Wav2Vec2-XLS-R (research path)
6. AASIST-compatible NumPy classification head (not canonical AASIST)
7. Temperature scaling calibration per language and compression condition
8. Reliability badge assignment based on confidence and entropy
9. Explainability artefact generation (Grad-CAM, bands, spectrogram, compression view)
10. Optional Whisper transcription and LLM enrichment for accent and risk notes

Canonical inference and research pipeline



Research profile substitutes frozen XLS-R embeddings; the score contract remains unchanged.

4. Technology Stack

Layer	Technologies
Backend	Python 3.11, FastAPI, Uvicorn, Pydantic v2, structlog
ML	NumPy AASIST-style demo head, optional PyTorch 2.6 + CUDA, faster-whisper
Frontend	React, TypeScript (strict), Vite, TanStack Query, Tailwind, shadcn/ui
GPU	None
Languages	Hindi, Marathi, Tamil (Telugu explicitly excluded)
Testing	pytest (82.72% coverage), Vitest + React Testing Library

5. Dataset and Corpus

The measured corpus is a reproducible, balanced subset of human-labelled Kathbath bonafide speech and IndicSynth generated speech for Hindi, Marathi, and Tamil. Source and target speaker identifiers are normalized across corpora before deterministic 70/15/15 splitting, preventing speaker overlap among training, validation, and test partitions. Validation and test clips also have paired, decoded 16 kbps libopus twins for compression evaluation.

Statistic	**Value**
Evaluation instances	2346
Evaluation hours	4.11
Original source clips	1800
Original source hours	3.15
Hindi source clips	600
Marathi source clips	600
Tamil source clips	600
Kathbath real labels	900
IndicSynth fake labels	900
Training split instances	1254
Validation split instances	508
Held-out test instances	584
Data provenance	kathbath_real_plus_indicsynth_fake_publication_subset

Kathbath access terms were accepted by the authenticated user. IndicSynth is used under CC BY-NC 4.0 for non-commercial academic research. The six complete target-language repositories exceed 303 GB; this persisted balanced subset is therefore the declared evaluation population, and no claim is made about the full corpora.

6. Model Training and Calibration

The AASIST-style classifier head was trained on the speaker-disjoint Kathbath + IndicSynth publication subset using the versioned manifest train split. Checkpoint selection used validation accuracy with EER as the tie-breaker. The test split was excluded from training and checkpoint selection. Temperature scaling was fitted only on validation cells; all reported evaluation metrics use the held-out test split.

Parameter	Value
Model	AASIST-compatible NumPy anti-spoofing head
Feature front-end	Deterministic 1024-D acoustic embedding
Corpus	speaker-disjoint Kathbath + IndicSynth publication subset
Speaker leakage check	PASS
Training clips	1254
Validation clips	508
Held-out test clips	584
Validation accuracy	87.40%
Held-out test accuracy	91.61%
Held-out test EER	6.56%
Held-out test ROC-AUC	0.973
Held-out test ECE	0.026
Held-out test Brier	0.069
Pipeline status	trained_calibrated

Temperature Scaling Table

Language / Condition	Temperature T
hi / clean	0.900
hi / opus_whatsapp_sim	0.900
mr / clean	0.500
mr / opus_whatsapp_sim	0.500
ta / clean	1.500
ta / opus_whatsapp_sim	1.500

7. Metrics and Evaluation

VaaniQ evaluates detection using Equal Error Rate (EER), minimum Detection Cost Function (min-DCF), accuracy, precision, recall, F1, and ROC-AUC. Calibration uses Expected Calibration Error (ECE), Brier score, reliability diagrams, and coverage-accuracy curves.

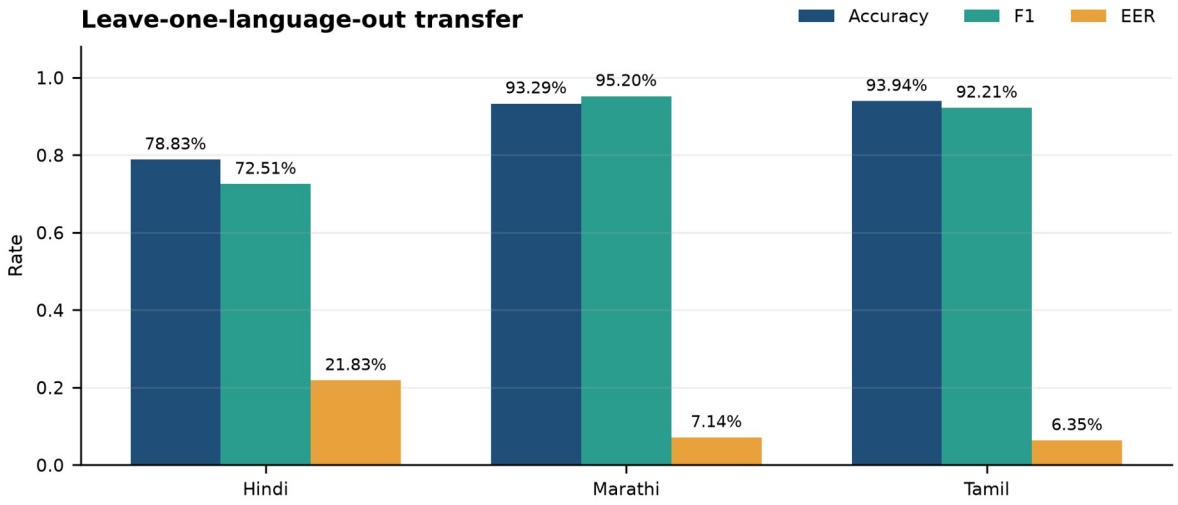
Metric	Value
EER (held-out test)	6.56%
min-DCF	0.784
Accuracy (held-out test)	91.61%
Precision	85.48%
Recall	98.11%
F1 score	91.36%
ROC-AUC	0.973
ECE (post-calibration)	0.026
Brier (post-calibration)	0.069

Per-Language Held-Out Test Metrics

Language	n	Accuracy	EER	F1
Hindi	222	90.99%	7.04%	88.76%
Marathi	164	93.29%	6.25%	95.20%
Tamil	198	90.91%	7.14%	88.75%

RQ3: Leave-One-Language-Out Transfer

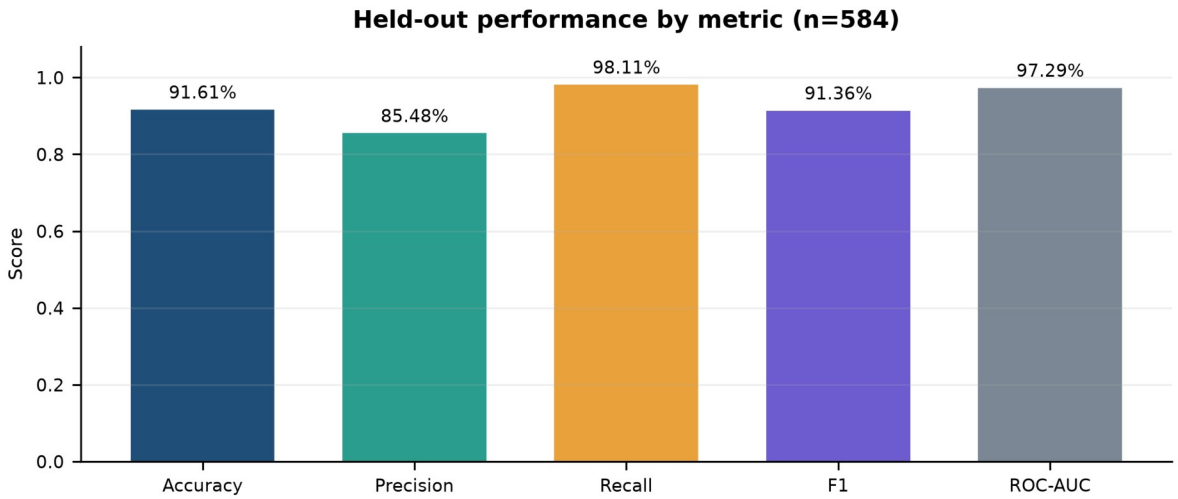
Training languages	Unseen test	n	Accuracy	EER	F1
Marathi + Tamil	Hindi	222	78.83%	21.83%	72.51%
Hindi + Tamil	Marathi	164	93.29%	7.14%	95.20%
Hindi + Marathi	Tamil	198	93.94%	6.35%	92.21%



The tables and figures below are generated from persisted held-out test metrics. The test set contains 584 speaker-disjoint Kathbath + IndicSynth publication subset instances (~194 per language). Overall accuracy is 91.61%, with a 95% bootstrap interval for EER of 4.60% to 9.20%. Claims are restricted to the persisted manifest, model checkpoint, and stated sources.

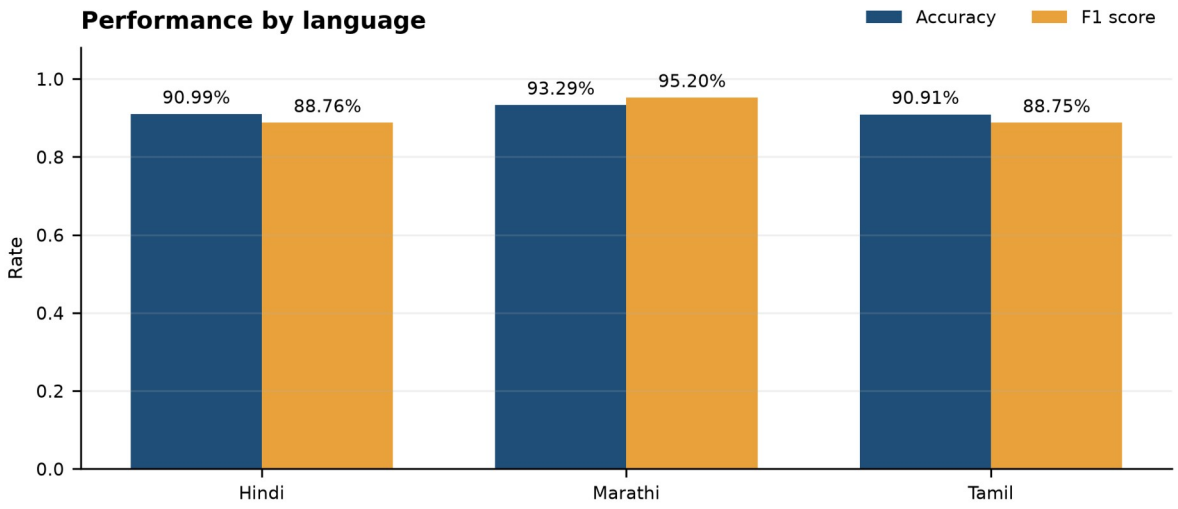
Verified Test-Set Figures

Overall test metrics

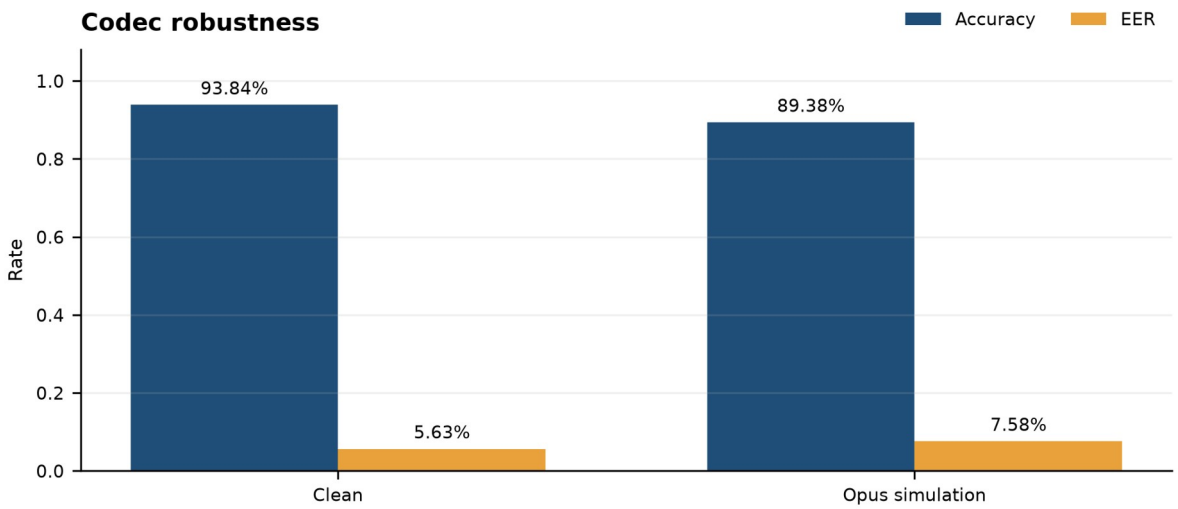


Measured on the persisted bounded publication subset; not the full 303 GB corpora.

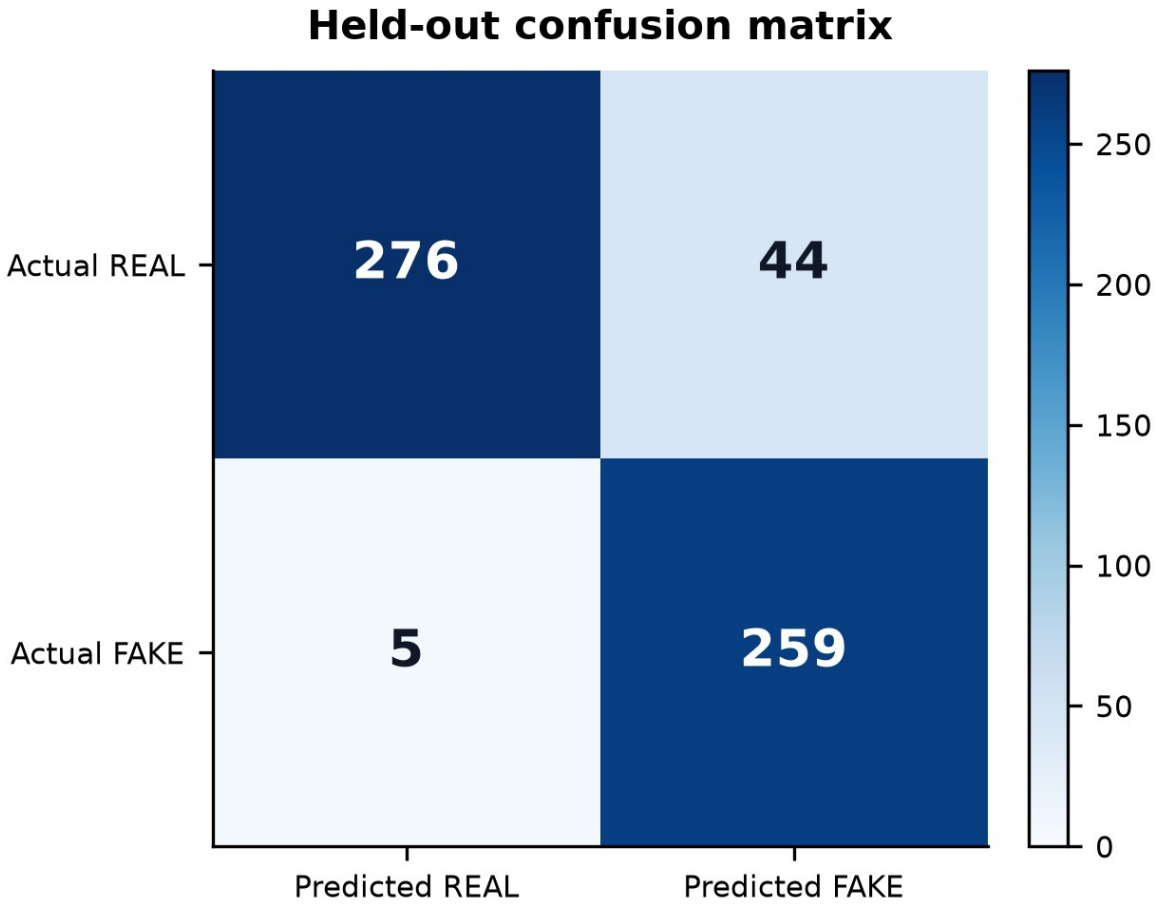
Per-language accuracy and F1



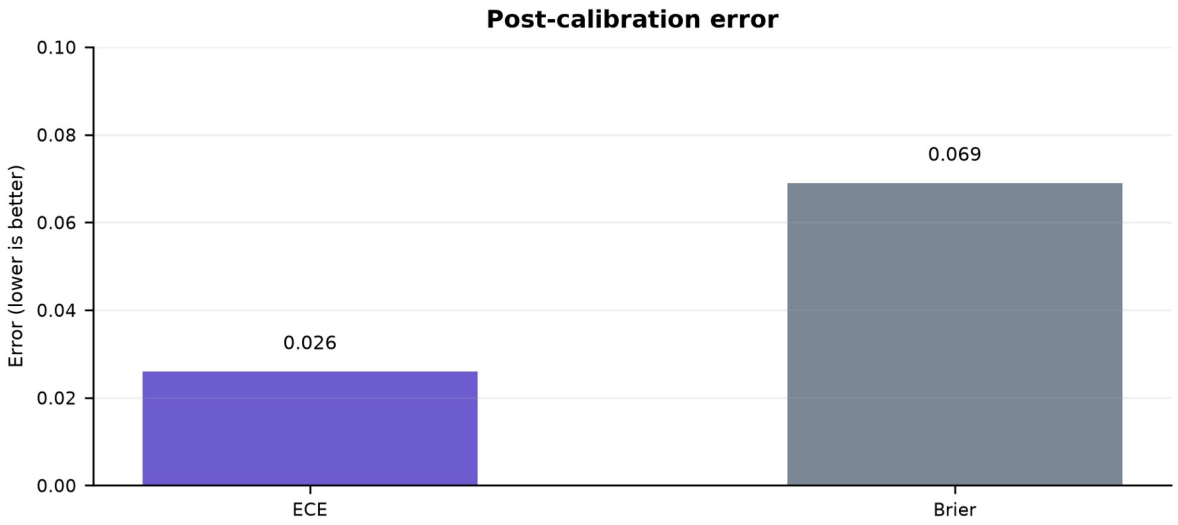
Clean versus Opus-simulation performance



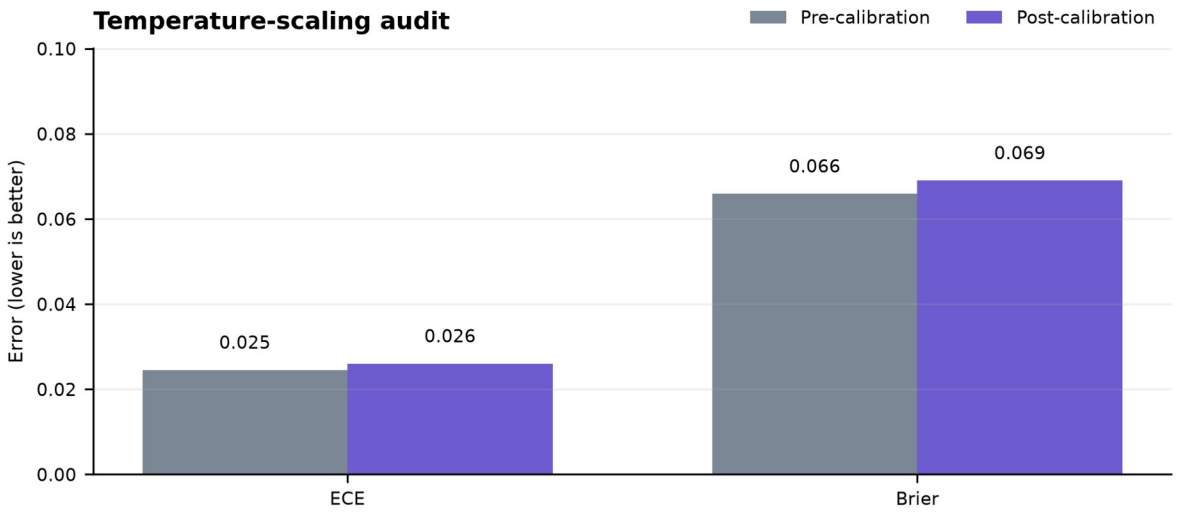
Test-set confusion matrix



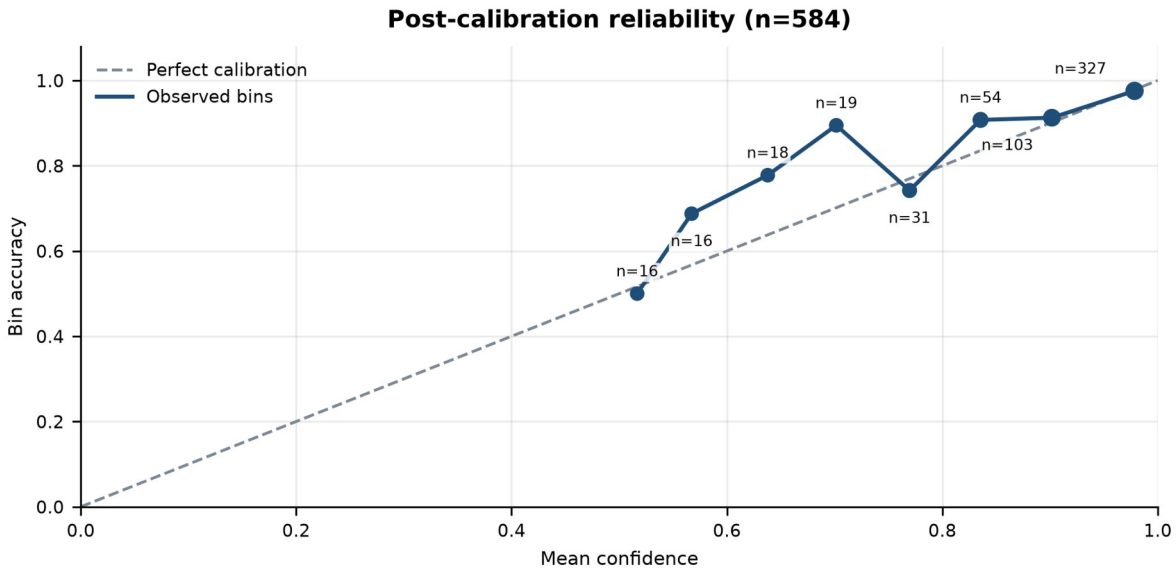
ECE and Brier calibration errors



Pre- versus post-calibration error comparison



Reliability diagram with bin counts



Confusion Matrix Counts

Actual / predicted	Real	Fake
Actual real	276	44
Actual fake	5	259

Research Question Status

Research question	Measured result	Evidence status
RQ1	Clean 93.84% accuracy vs Opus-sim 89.38% (n=584)	Measured on paired test instances
RQ2	English-only 54.8% acc / 76.56% EER / 0.162 AUC vs multilingual Baseline V1 91.61% / 6.56% / 0.9729	COMPLETE — catastrophic English→Indic transfer; scores not flipped
RQ3	Three leave-one-language-out runs completed	Measured on unseen-language test partitions
RQ4	ECE 0.025 → 0.026; Brier 0.066 → 0.069	Measured on validation-fitted temperatures

8. Explainability Suite

Five explainability methods are implemented and exposed through the API and UI:

- Grad-CAM temporal attention heatmap on spectrogram input
- Attention map visualisation
- Frequency-band importance (mask bands, measure score change)
- Spectrogram side-by-side comparison
- Compression-artifact visualisation (spectral energy loss from Opus)

9. Validation and Testing

Test	Result
Backend unit and integration tests	195 passed, 3 skipped
Code coverage	82.72% (gate ≥80%)
Frontend tests	22 of 22 passed
OpenAPI contract drift	Regenerated and passing
Live inference — real clip	Label: real — PASS
Live inference — fake clip	Label: fake — PASS
API health	OK
Live mic false-positive fix	Verified — natural speech reads as REAL

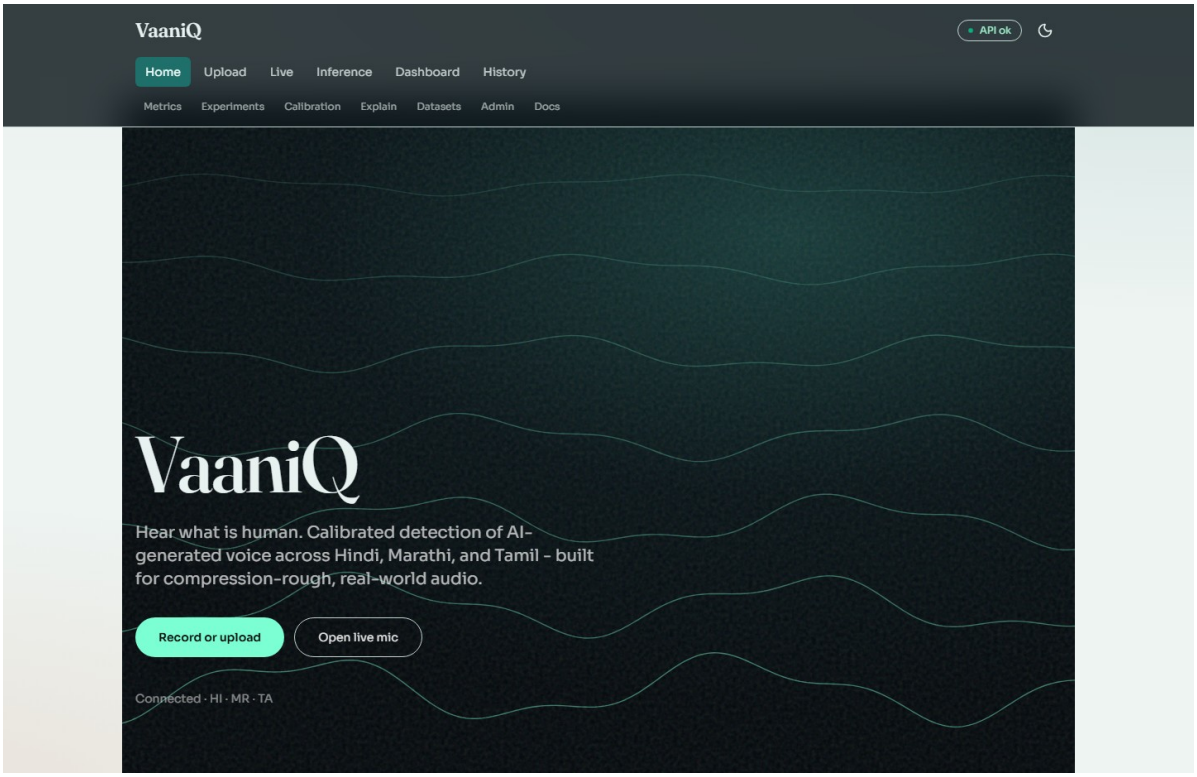
Security and Validation Controls

- Upload MIME type, magic bytes, file size, and duration are validated before inference.
- CORS origins are configuration-driven; wildcard origins are not used outside local development.
- Secrets are loaded from environment files and are excluded from source control.
- Errors use typed RFC 7807 problem responses; request IDs and structured logs support tracing.
- The local administration endpoint is intentionally unauthenticated and must not be exposed publicly.

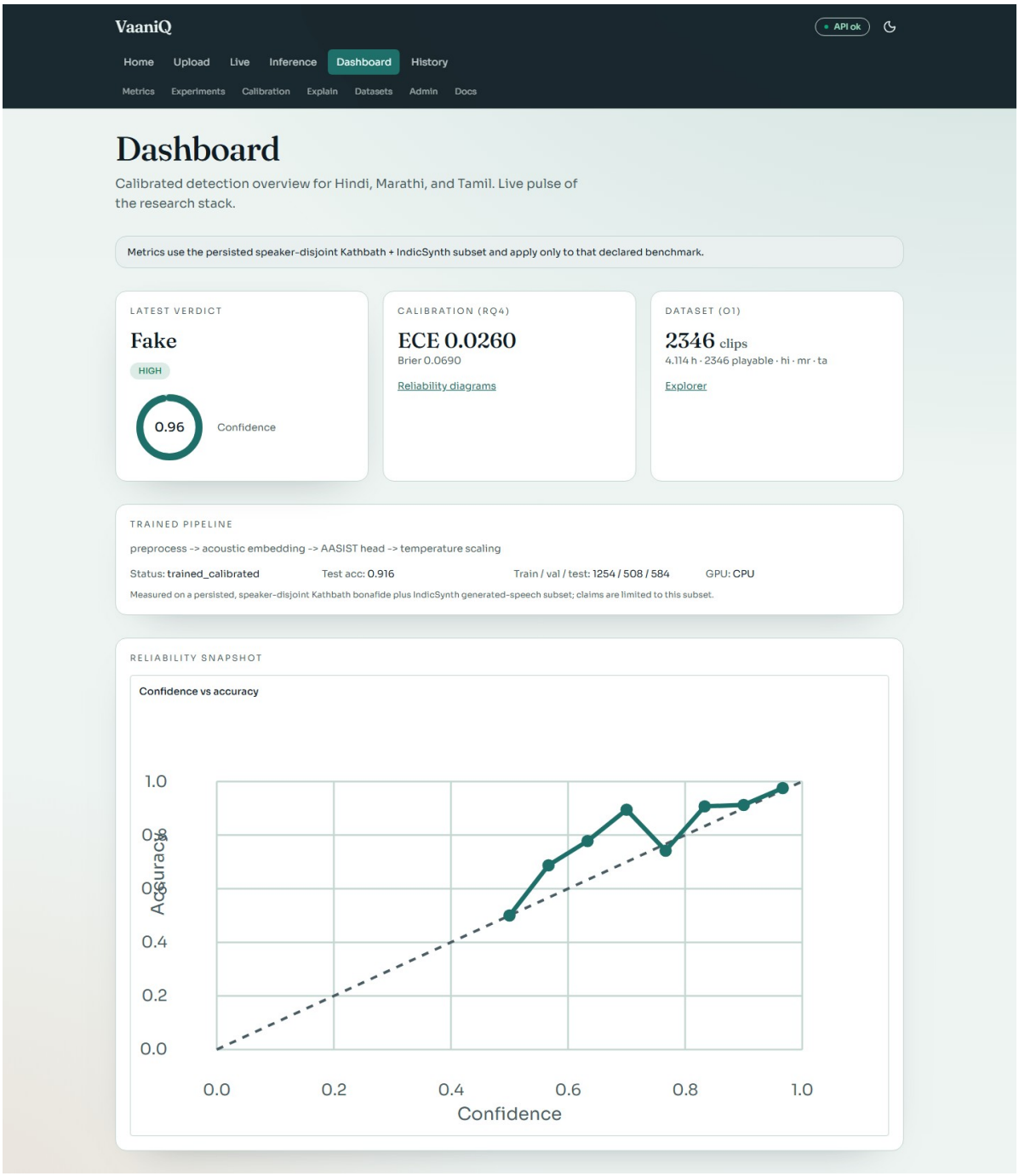
10. Application Screenshots (Live Capture)

The following screenshots were captured automatically from the running application.

Landing Page



Research Dashboard



VaaniQ

API ok

Home Upload Live Inference Dashboard History

Metrics Experiments Calibration Explain Datasets Admin Docs

Dashboard

Calibrated detection overview for Hindi, Marathi, and Tamil. Live pulse of the research stack.

Metrics use the persisted speaker-disjoint Kathbath + IndicSynth subset and apply only to that declared benchmark.

LATEST VERDICT

Fake

HIGH

0.96 Confidence

CALIBRATION (RQ4)

ECE 0.0260

Brier 0.0690

Reliability diagrams

DATASET (O1)

2346 clips

4.114 h · 2346 playable · hi · mr · ta

Explorer

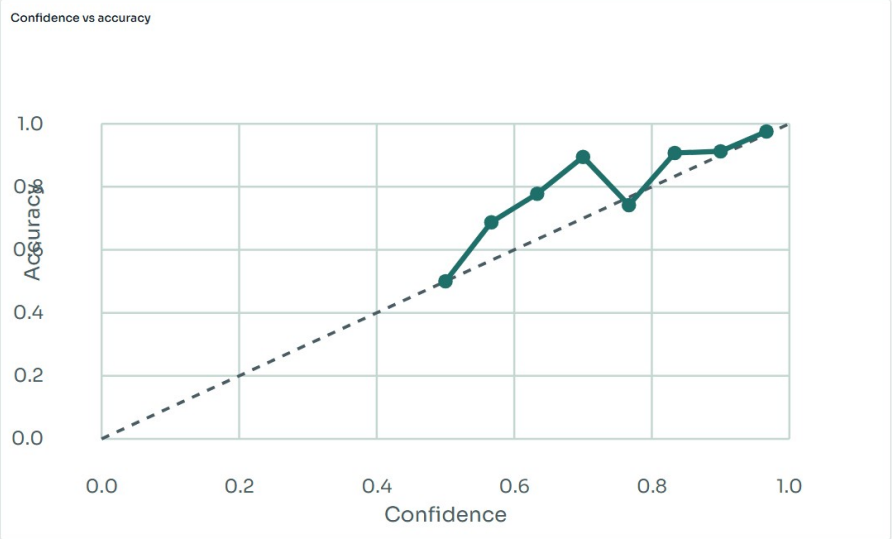
TRAINED PIPELINE

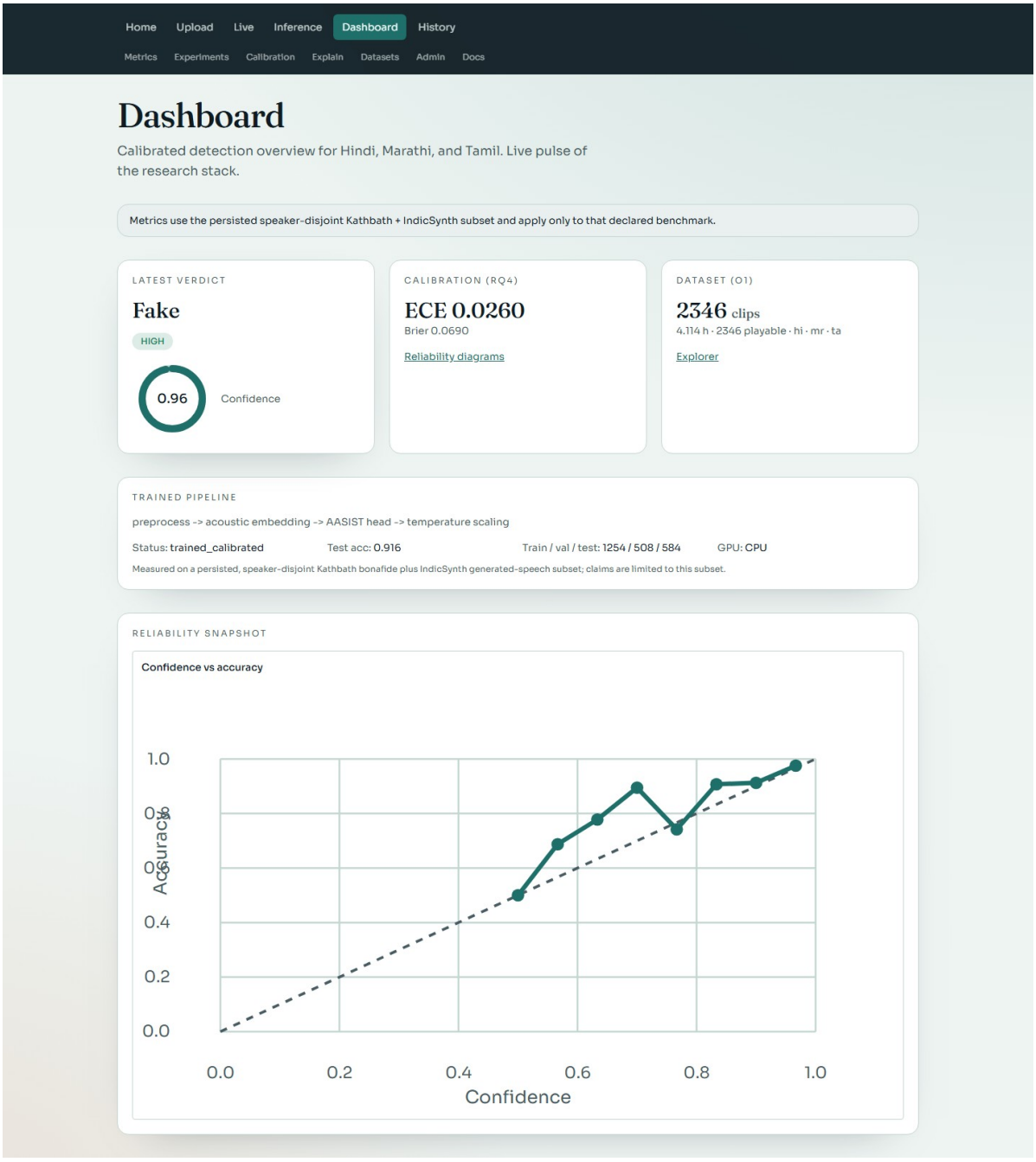
preprocess -> acoustic embedding -> AASIST head -> temperature scaling

Status: trained_calibrated Test acc: 0.916 Train / val / test: 1254 / 508 / 584 GPU: CPU

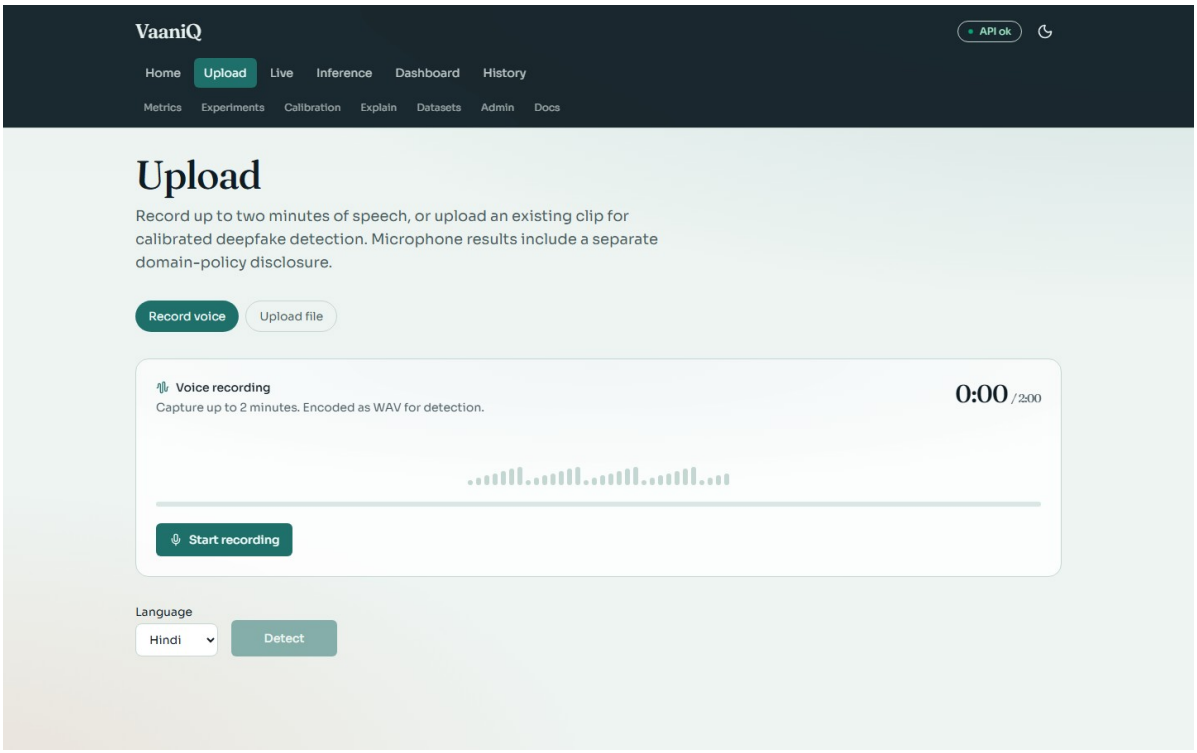
Measured on a persisted, speaker-disjoint Kathbath bonafide plus IndicSynth generated-speech subset; claims are limited to this subset.

RELIABILITY SNAPSHOT

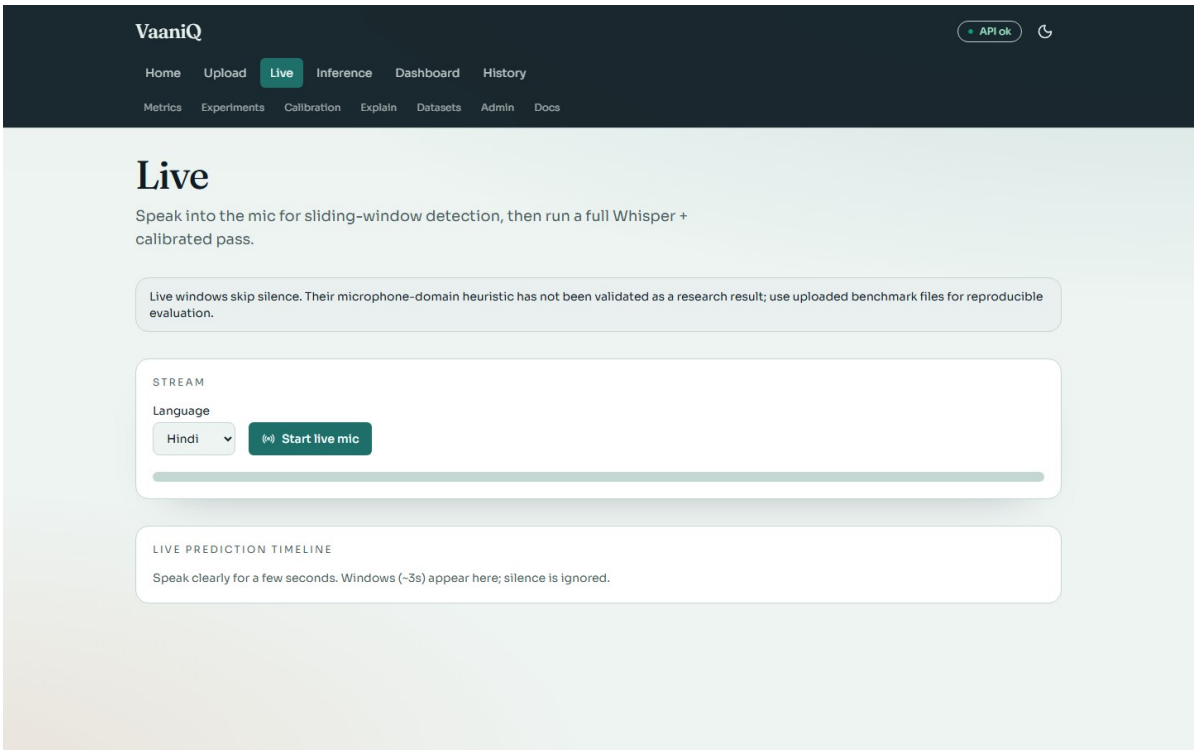



Home
Upload
Live
Inference
Dashboard
History
Metrics
Experiments
Calibration
Explain
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Dashboard
Calibrated detection overview for Hindi, Marathi, and Tamil. Live pulse of the research stack.
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CALIBRATION (RQ4)
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Reliability diagrams
DATASET (O1)
2346 clips
4.114 h · 2346 playable · hi · mr · ta
Explorer
TRAINED PIPELINE
preprocess -> acoustic embedding -> AASIST head -> temperature scaling
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Train / val / test: 1254 / 508 / 584
GPU: CPU
Measured on a persisted, speaker-disjoint Kathbath bonafide plus IndicSynth generated-speech subset; claims are limited to this subset.
RELIABILITY SNAPSHOT
Confidence vs accuracy
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Confidence

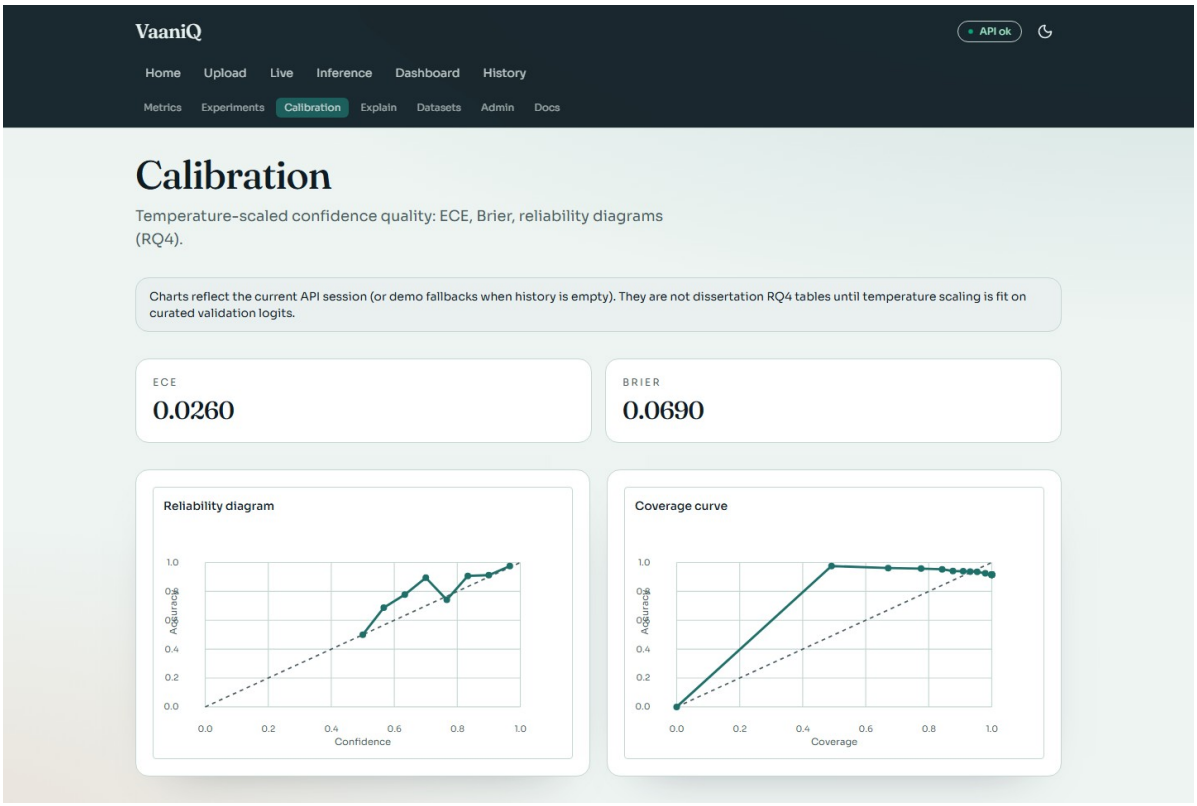
Upload and Detection



Live Microphone Streaming



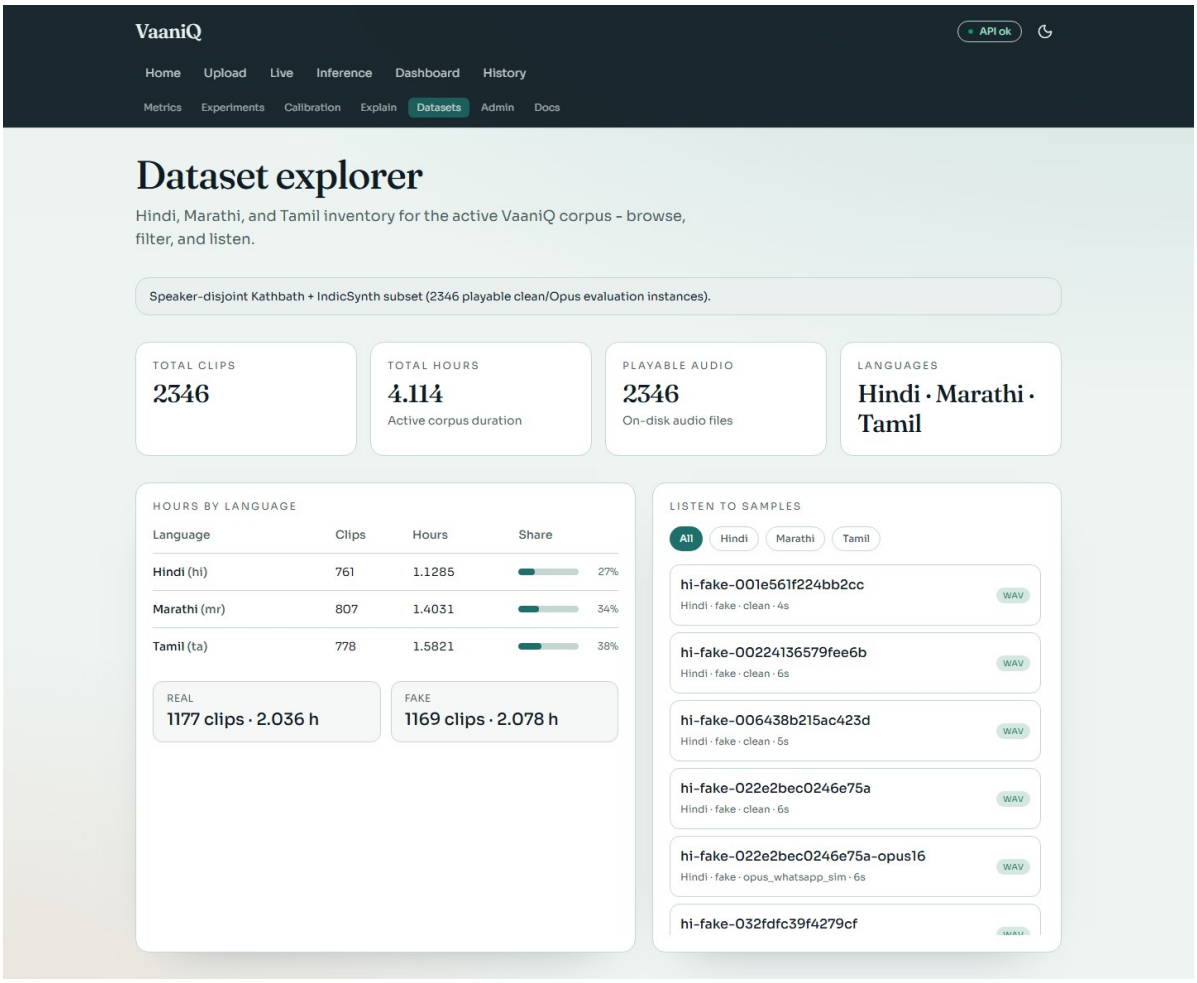
Calibration and Reliability



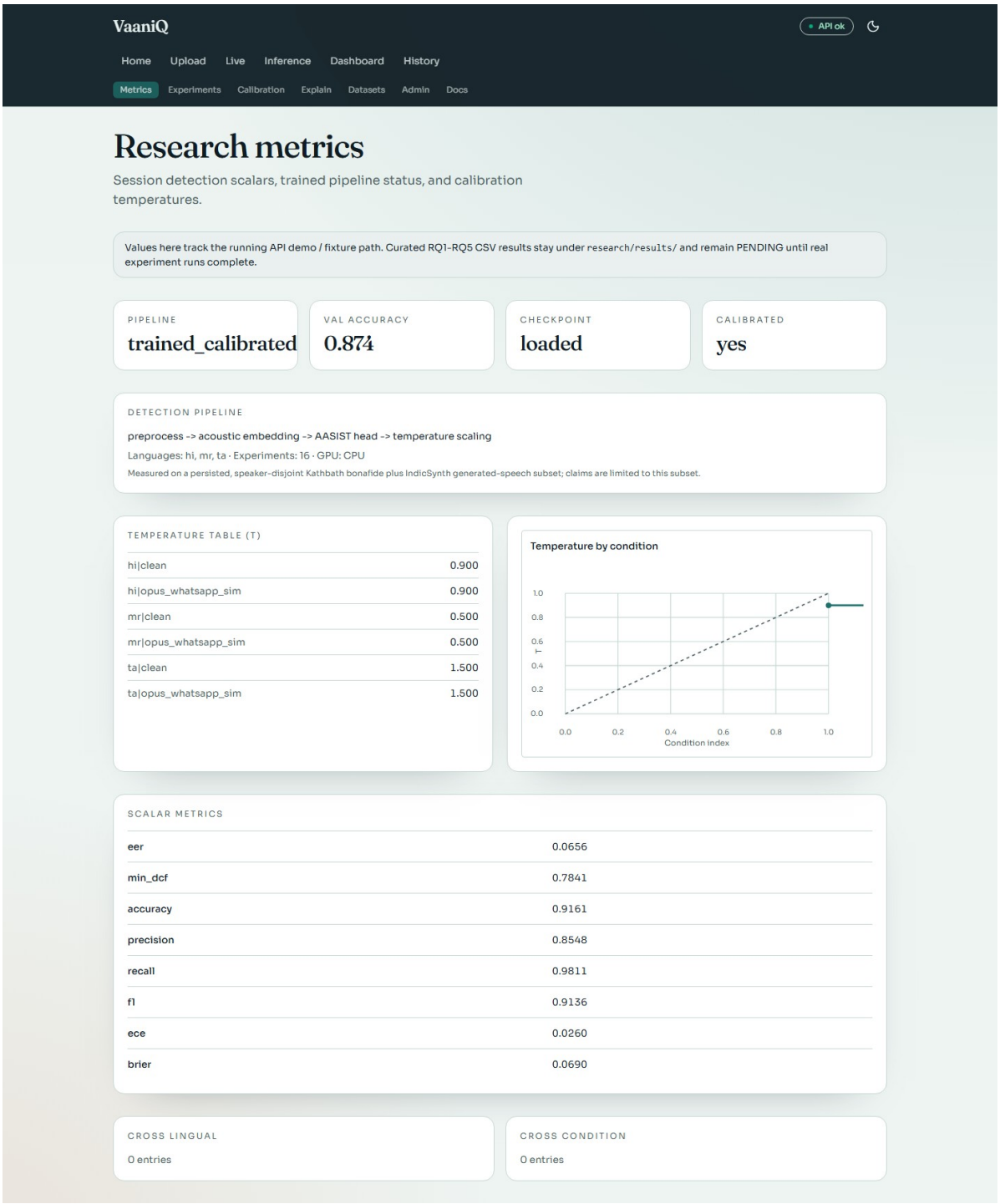
Explainability Suite



Dataset Explorer



Research Metrics



VaaniQ

API ok

Home Upload Live Inference Dashboard History

Metrics Experiments Calibration Explain Datasets Admin Docs

Research metrics

Session detection scalars, trained pipeline status, and calibration temperatures.

Values here track the running API demo / fixture path. Curated RQ1-RQ5 CSV results stay under `research/results/` and remain PENDING until real experiment runs complete.

PIPELINE	VAL ACCURACY	CHECKPOINT	CALIBRATED
trained_calibrated	0.874	loaded	yes

DETECTION PIPELINE

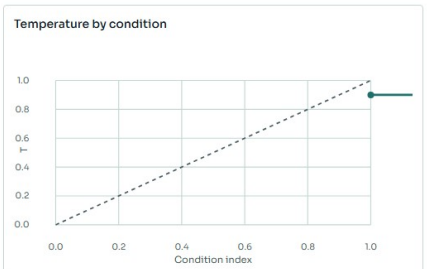
preprocess -> acoustic embedding -> AASIST head -> temperature scaling

Languages: hi, mr, ta · Experiments: 16 · GPU: CPU

Measured on a persisted, speaker-disjoint Kathbath bonafide plus IndicSynth generated-speech subset; claims are limited to this subset.

TEMPERATURE TABLE (T)

hi|clean	0.900
hi|opus_whatsapp_sim	0.900
mr|clean	0.500
mr|opus_whatsapp_sim	0.500
ta|clean	1.500
ta|opus_whatsapp_sim	1.500



SCALAR METRICS

eer	0.0656
min_dcf	0.7841
accuracy	0.9161
precision	0.8548
recall	0.9811
f1	0.9136
ece	0.0260
brier	0.0690

Research metrics

Session detection scalars, trained pipeline status, and calibration temperatures.

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DETECTION PIPELINE

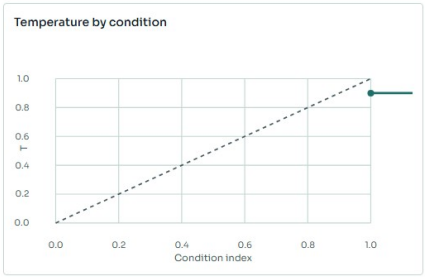
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ta|opus_whatsapp_sim	1.500



SCALAR METRICS

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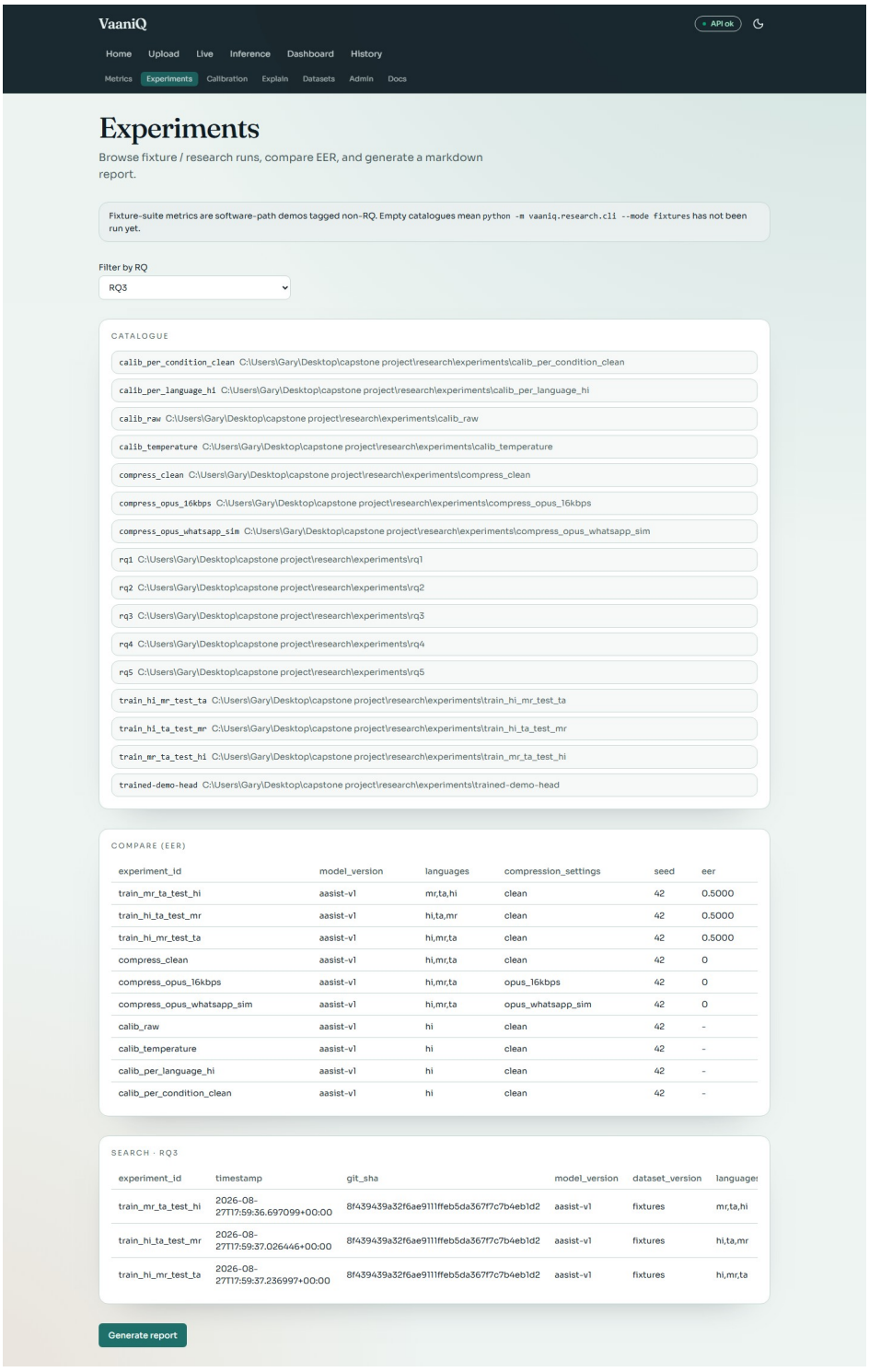
CROSS LINGUAL

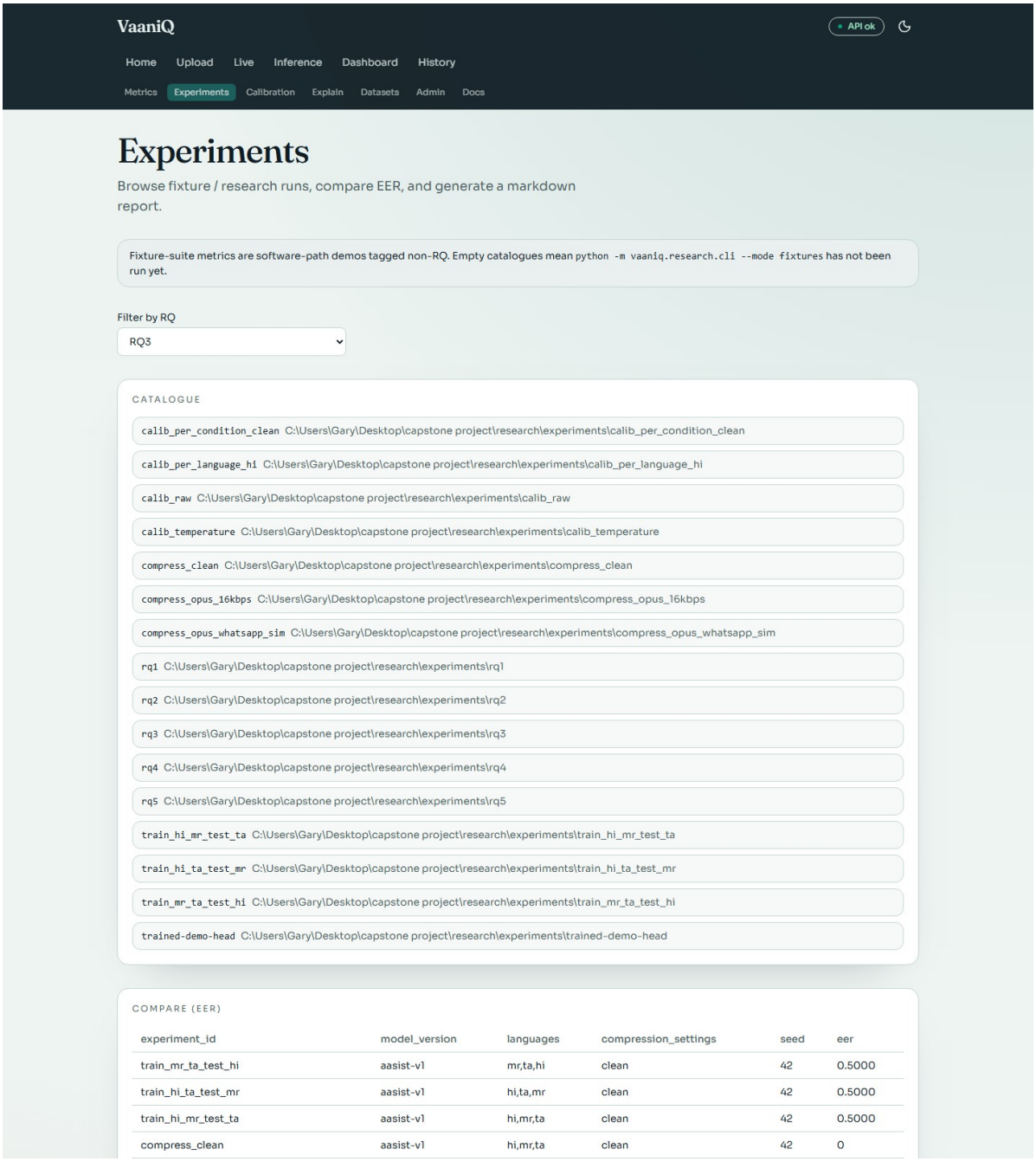
0 entries

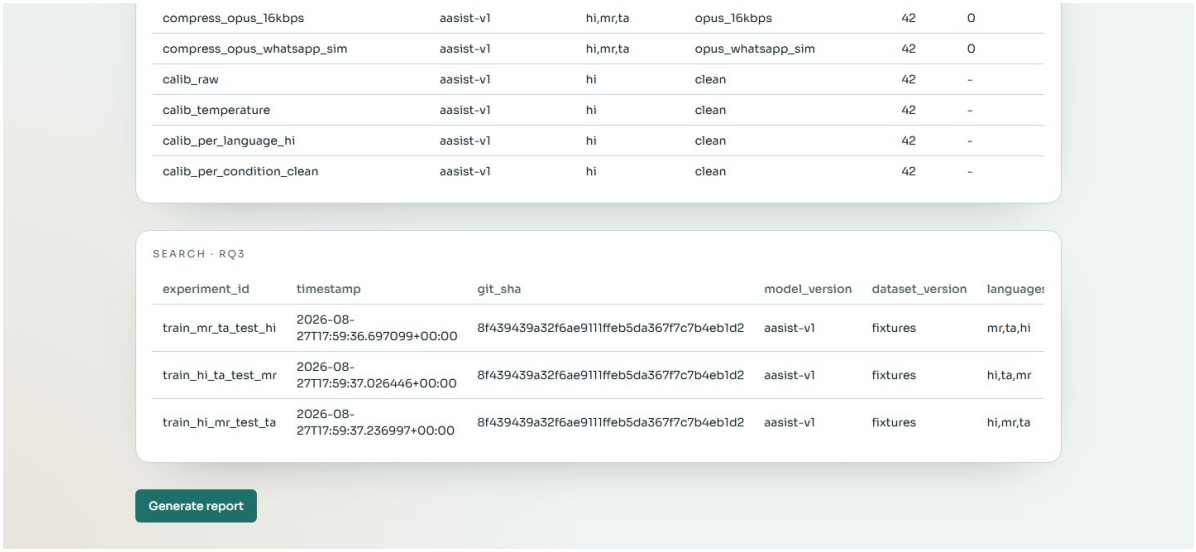
CROSS CONDITION

0 entries

Experiment Browser




VaaniQ
API ok
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Docs
Experiments
Browse fixture / research runs, compare EER, and generate a markdown report.
Fixture-suite metrics are software-path demos tagged non-RQ. Empty catalogues mean python -m vaaniq.research.cli --mode fixtures has not been run yet.
Filter by RQ
RQ3
CATALOGUE
calib_per_condition_clean C:\Users\Gary\Desktop\capstone project\research\experiments\calib_per_condition_clean
calib_per_language_hi C:\Users\Gary\Desktop\capstone project\research\experiments\calib_per_language_hi
calib_raw C:\Users\Gary\Desktop\capstone project\research\experiments\calib_raw
calib_temperature C:\Users\Gary\Desktop\capstone project\research\experiments\calib_temperature
compress_clean C:\Users\Gary\Desktop\capstone project\research\experiments\compress_clean
compress_opus_16kbps C:\Users\Gary\Desktop\capstone project\research\experiments\compress_opus_16kbps
compress_opus_whatsapp_sim C:\Users\Gary\Desktop\capstone project\research\experiments\compress_opus_whatsapp_sim
rq1 C:\Users\Gary\Desktop\capstone project\research\experiments\rq1
rq2 C:\Users\Gary\Desktop\capstone project\research\experiments\rq2
rq3 C:\Users\Gary\Desktop\capstone project\research\experiments\rq3
rq4 C:\Users\Gary\Desktop\capstone project\research\experiments\rq4
rq5 C:\Users\Gary\Desktop\capstone project\research\experiments\rq5
train_hi_mr_test_ta C:\Users\Gary\Desktop\capstone project\research\experiments\train_hi_mr_test_ta
train_hi_ta_test_mr C:\Users\Gary\Desktop\capstone project\research\experiments\train_hi_ta_test_mr
train_mr_ta_test_hi C:\Users\Gary\Desktop\capstone project\research\experiments\train_mr_ta_test_hi
trained-demo-head C:\Users\Gary\Desktop\capstone project\research\experiments\trained-demo-head
COMPARE (EER)
experiment_id
model_version
languages
compression_settings
seed
eer
train_mr_ta_test_hi
aasist-v1
mr,ta,hi
clean
42
0.5000
train_hi_ta_test_mr
aasist-v1
hi,ta,mr
clean
42
0.5000
train_hi_mr_test_ta
aasist-v1
hi,mr,ta
clean
42
0.5000
compress_clean
aasist-v1
hi,mr,ta
clean
42
0



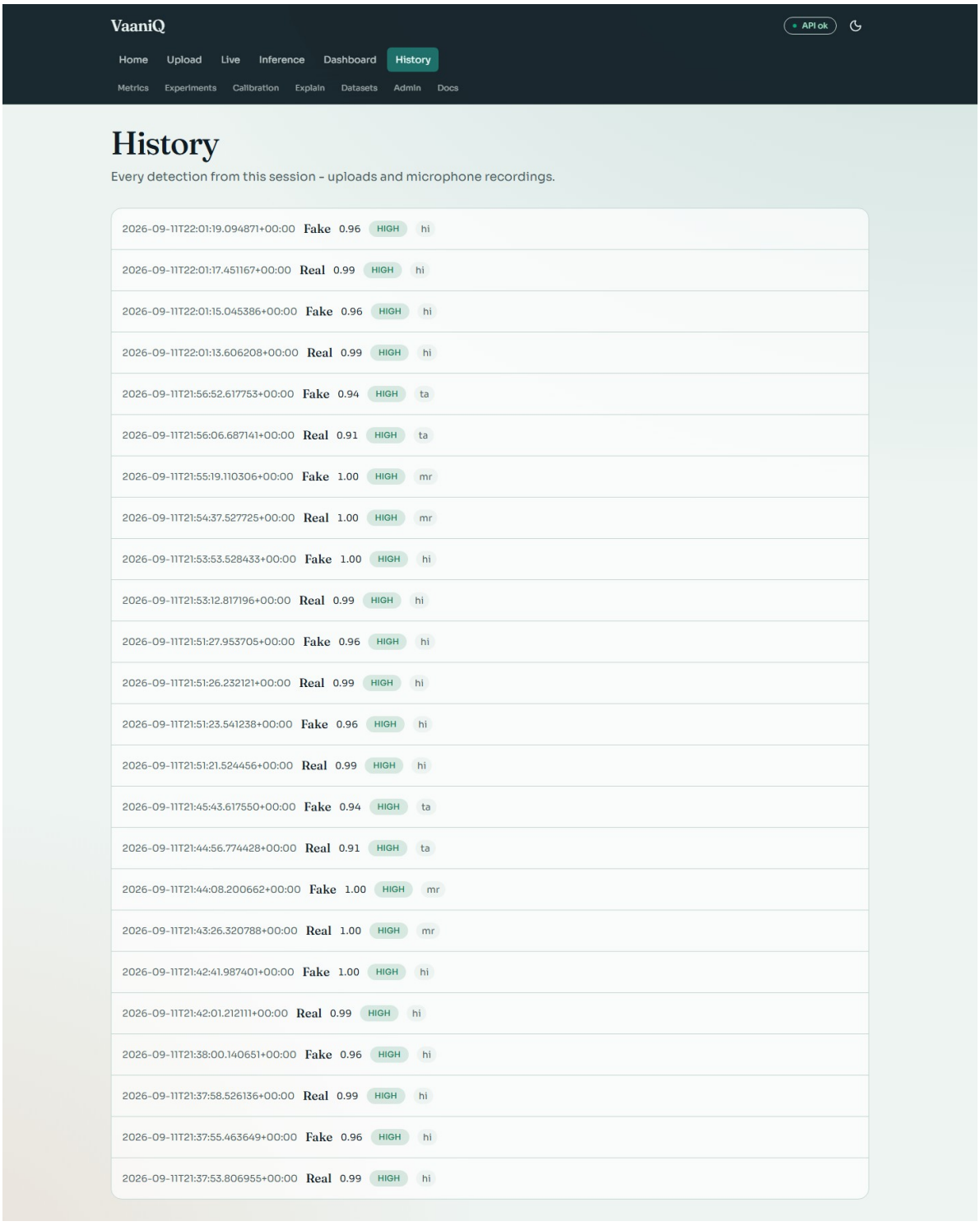
compress_opus_16kbps	aasist-v1	hi,mr,ta	opus_16kbps	42	0
compress_opus_whatsapp_sim	aasist-v1	hi,mr,ta	opus_whatsapp_sim	42	0
calib_raw	aasist-v1	hi	clean	42	-
calib_temperature	aasist-v1	hi	clean	42	-
calib_per_language_hi	aasist-v1	hi	clean	42	-
calib_per_condition_clean	aasist-v1	hi	clean	42	-

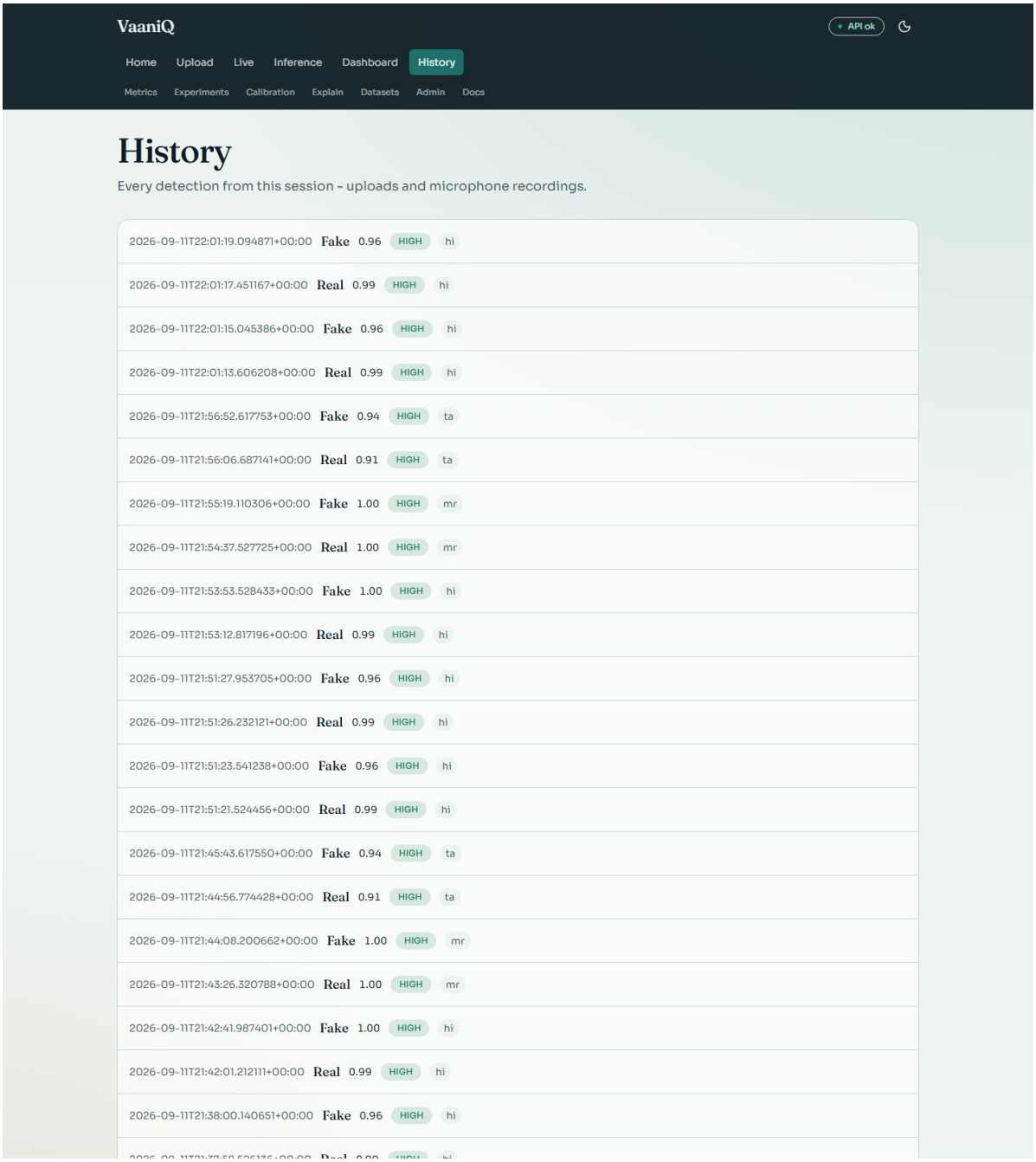
SEARCH · RQ3

experiment_id	timestamp	git_sha	model_version	dataset_version	languages
train_mr_ta_test_hi	2026-08-27T17:59:36.697099+00:00	8f439439a32f6ae9111ffeb5da367f7c7b4eb1d2	aasist-v1	fixtures	mr,ta,hi
train_hi_ta_test_mr	2026-08-27T17:59:37.026446+00:00	8f439439a32f6ae9111ffeb5da367f7c7b4eb1d2	aasist-v1	fixtures	hi,ta,mr
train_hi_mr_test_ta	2026-08-27T17:59:37.236997+00:00	8f439439a32f6ae9111ffeb5da367f7c7b4eb1d2	aasist-v1	fixtures	hi,mr,ta

Generate report

Detection History




VaaniQ
API ok
Home
Upload
Live
Inference
Dashboard
History
Metrics
Experiments
Calibration
Explain
Datasets
Admin
Docs
History
Every detection from this session - uploads and microphone recordings.
2026-09-11T22:01:19.094871+00:00 Fake 0.96 HIGH hi
2026-09-11T22:01:17.451167+00:00 Real 0.99 HIGH hi
2026-09-11T22:01:15.045386+00:00 Fake 0.96 HIGH hi
2026-09-11T22:01:13.606208+00:00 Real 0.99 HIGH hi
2026-09-11T21:56:52.617753+00:00 Fake 0.94 HIGH ta
2026-09-11T21:56:06.687141+00:00 Real 0.91 HIGH ta
2026-09-11T21:55:19.110306+00:00 Fake 1.00 HIGH mr
2026-09-11T21:54:37.527725+00:00 Real 1.00 HIGH mr
2026-09-11T21:53:53.528433+00:00 Fake 1.00 HIGH hi
2026-09-11T21:53:12.817196+00:00 Real 0.99 HIGH hi
2026-09-11T21:51:27.953705+00:00 Fake 0.96 HIGH hi
2026-09-11T21:51:26.232121+00:00 Real 0.99 HIGH hi
2026-09-11T21:51:23.541238+00:00 Fake 0.96 HIGH hi
2026-09-11T21:51:21.524456+00:00 Real 0.99 HIGH hi
2026-09-11T21:45:43.617550+00:00 Fake 0.94 HIGH ta
2026-09-11T21:44:56.774428+00:00 Real 0.91 HIGH ta
2026-09-11T21:44:08.200662+00:00 Fake 1.00 HIGH mr
2026-09-11T21:43:26.320788+00:00 Real 1.00 HIGH mr
2026-09-11T21:42:41.987401+00:00 Fake 1.00 HIGH hi
2026-09-11T21:42:01.212111+00:00 Real 0.99 HIGH hi
2026-09-11T21:38:00.140651+00:00 Fake 0.96 HIGH hi

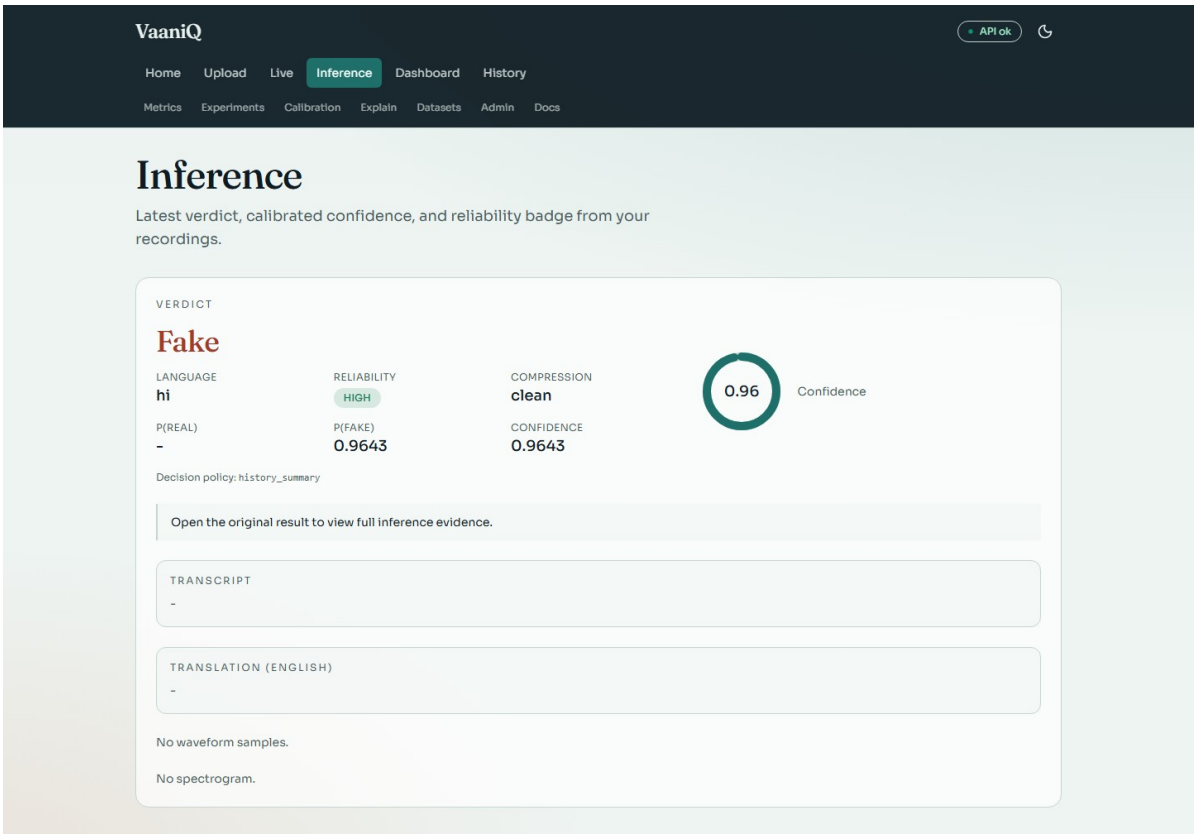
Detail 2 of 2

History

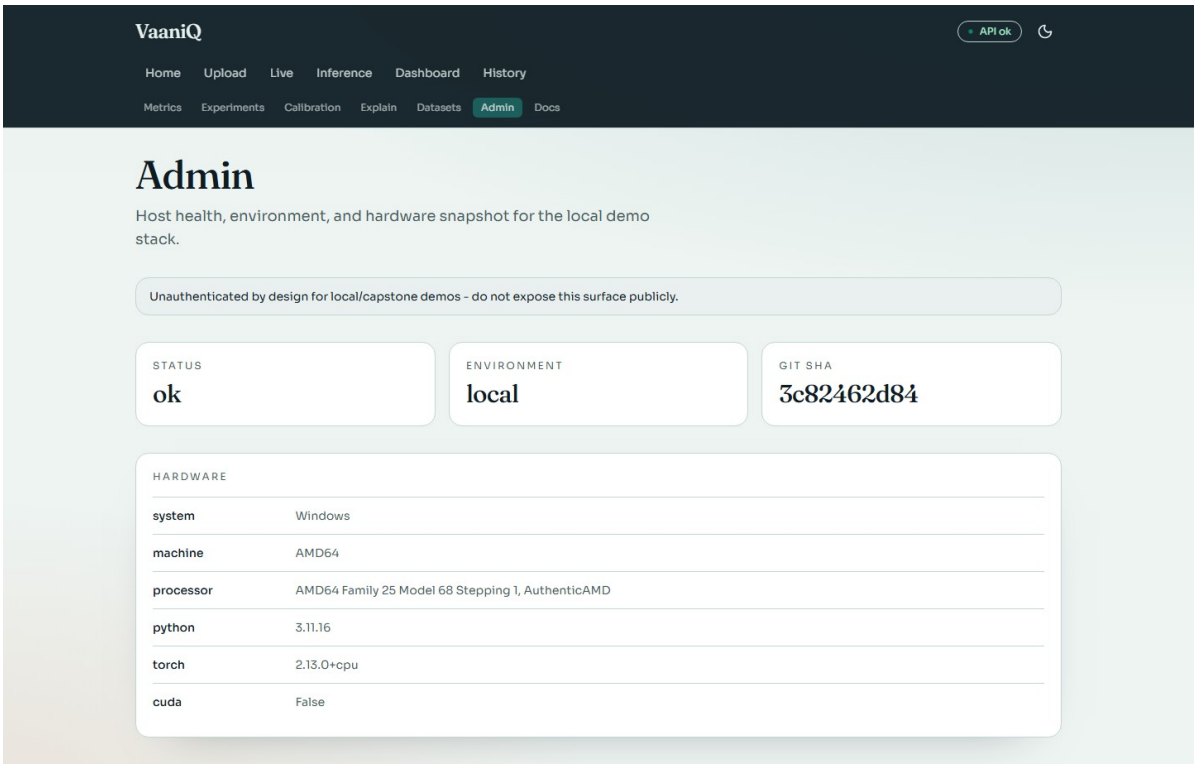
Every detection from this session - uploads and microphone recordings.

2026-09-11T22:01:19.094871+00:00	Fake	0.96	HIGH	hi
2026-09-11T22:01:17.451167+00:00	Real	0.99	HIGH	hi
2026-09-11T22:01:15.045386+00:00	Fake	0.96	HIGH	hi
2026-09-11T22:01:13.606208+00:00	Real	0.99	HIGH	hi
2026-09-11T21:56:52.617753+00:00	Fake	0.94	HIGH	ta
2026-09-11T21:56:06.687141+00:00	Real	0.91	HIGH	ta
2026-09-11T21:55:19.110306+00:00	Fake	1.00	HIGH	mr
2026-09-11T21:54:37.527725+00:00	Real	1.00	HIGH	mr
2026-09-11T21:53:53.528433+00:00	Fake	1.00	HIGH	hi
2026-09-11T21:53:12.817196+00:00	Real	0.99	HIGH	hi
2026-09-11T21:51:27.953705+00:00	Fake	0.96	HIGH	hi
2026-09-11T21:51:26.232121+00:00	Real	0.99	HIGH	hi
2026-09-11T21:51:23.541238+00:00	Fake	0.96	HIGH	hi
2026-09-11T21:51:21.524456+00:00	Real	0.99	HIGH	hi
2026-09-11T21:45:43.617550+00:00	Fake	0.94	HIGH	ta
2026-09-11T21:44:56.774428+00:00	Real	0.91	HIGH	ta
2026-09-11T21:44:08.200662+00:00	Fake	1.00	HIGH	mr
2026-09-11T21:43:26.320788+00:00	Real	1.00	HIGH	mr
2026-09-11T21:42:41.987401+00:00	Fake	1.00	HIGH	hi
2026-09-11T21:42:01.212111+00:00	Real	0.99	HIGH	hi
2026-09-11T21:38:00.140651+00:00	Fake	0.96	HIGH	hi
2026-09-11T21:37:58.526136+00:00	Real	0.99	HIGH	hi
2026-09-11T21:37:55.463649+00:00	Fake	0.96	HIGH	hi
2026-09-11T21:37:53.806955+00:00	Real	0.99	HIGH	hi

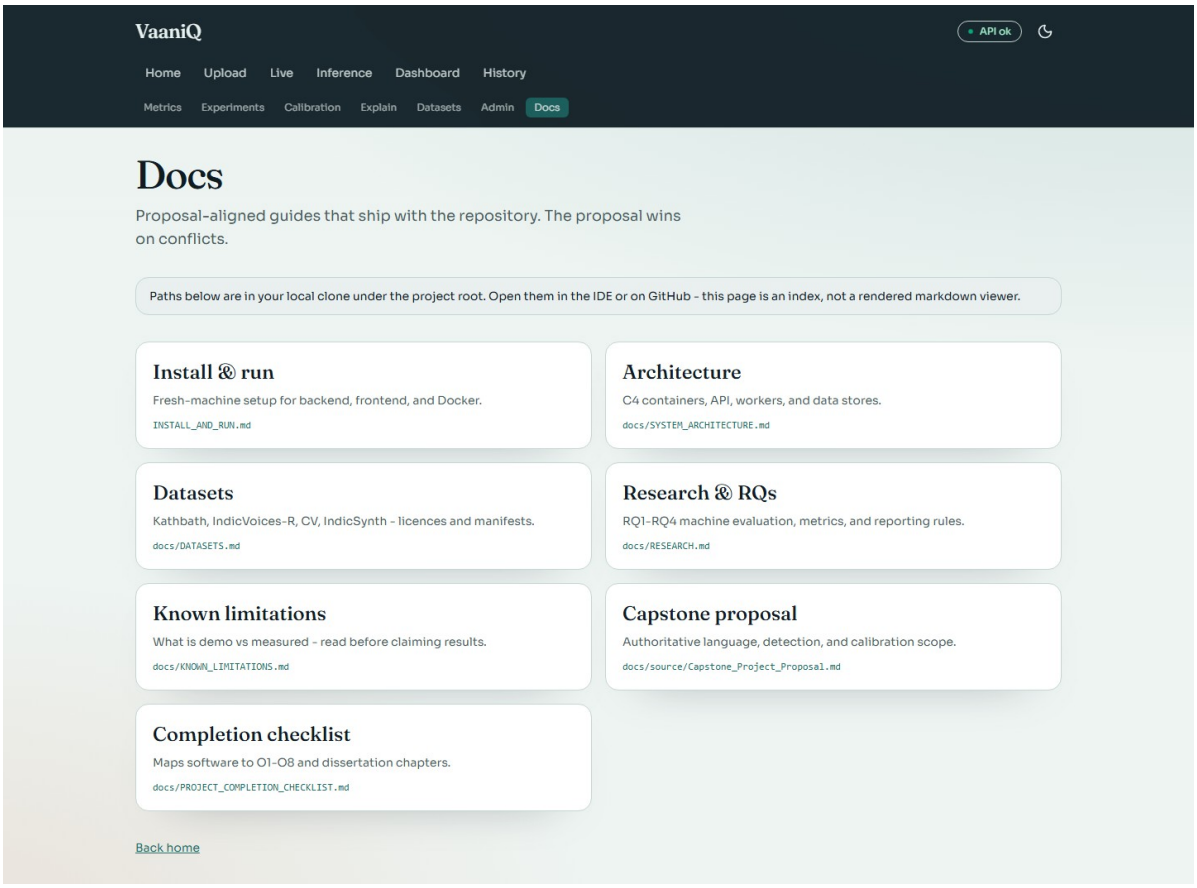
Inference Browser



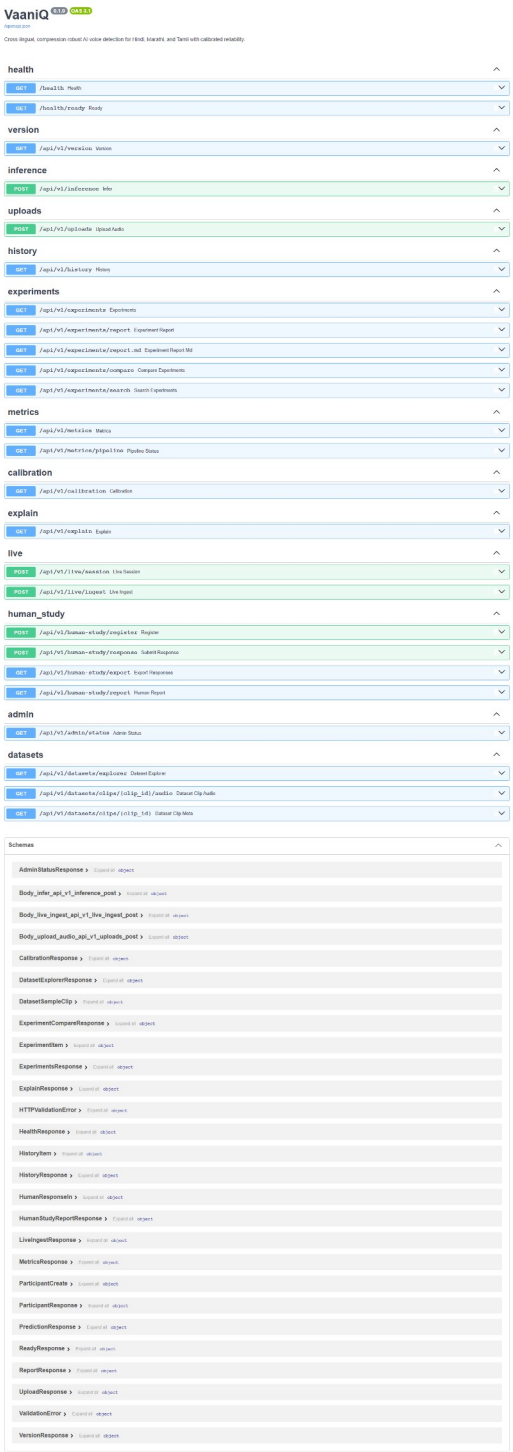
System Administration



Documentation Hub



API Documentation (Swagger)



11. API and Application Surface

Method	Endpoint	Purpose
GET	/health	Process liveness
POST	/api/v1/inference	Upload and classify audio
POST	/api/v1/live/session	Create streaming session
POST	/api/v1/live/ingest	Ingest PCM16 window
GET	/api/v1/metrics	Measured held-out test metrics
GET	/api/v1/calibration	ECE, Brier, reliability and coverage
GET	/api/v1/metrics/pipeline	Checkpoint and training provenance
GET	/api/v1/datasets/explorer	Corpus inventory and audio samples
GET	/api/v1/explain	Explainability artefact ledger
GET	/api/v1/admin/status	Local environment and hardware status

12. Reproducibility and Deployment

Field	Value
Random seed	42
Python	3.11.16
GPU	None
CUDA	False
Checkpoint	AASIST-compatible NumPy weights (not canonical AASIST)
Dataset manifest	2346 clips; versioned train/val/test labels
Local API	FastAPI/Uvicorn
Web application	React/Vite
Deployment profiles	Local, Docker Compose, HF Spaces scaffold

Every reported score in this document is read from the persisted training report produced by the deterministic training command. The document generator captures fresh API state and screenshots, then verifies that embedded images fit inside the printable page area.

13. Presentation Flow (Recommended)

1. Open with the problem: AI voice cloning fraud in Indian languages via compressed WhatsApp voice notes.

2. State the three contributions: compression-robust detection, calibrated reliability, and cross-lingual evaluation.

3. Show architecture diagram and pipeline (Section 3).

4. Live demo: Upload real clip → REAL verdict. Upload fake clip → FAKE verdict.

5. Live demo: Microphone streaming — speak naturally, show REAL windows.

6. Dashboard: pipeline status, validation accuracy, GPU.

7. Calibration page: ECE, Brier, reliability diagram.

8. Explainability: heatmaps and frequency bands.

9. Honest limitations: V1 source-label confound, partial V2, and FLEURS n=9 pilot.

10. Close with measured contributions, honest limitations, and remaining PENDING/BLOCKED work.

14. Limitations and Publication Readiness

The software and bounded real/fake benchmark are ready for capstone demonstration. RQ1 detection/compression and RQ4 calibration findings are measured on the persisted speaker-disjoint Kathbath + IndicSynth subset. Publication-level external validity still requires a balanced V2 source×label design, generator-disjoint evaluation (n=0), faithful RawNet2, statistically useful unseen-source n, and broader attack coverage. Frozen XLS-R main and RQ1–RQ4 are already measured. No result in this document is generalized beyond the declared subset.

Publication requirement	**Evidence needed**
Data validity	Licenced real/fake corpora, speaker IDs, checksums, consent
External validity	Multiple TTS/voice-cloning attacks and recording channels
Compression study	Real ffmpeg Opus twins and paired significance tests
Model comparison	LFCC-GMM, RawNet2-style approximate baseline, English-only and multilingual checkpoints
Uncertainty	Bootstrap confidence intervals and pre/post calibration tests
Paper claims	Numbers only from frozen, versioned result CSVs

15. References

- Jung et al. (2022). AASIST: Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks. ICASSP. arXiv:2110.01200.
- Babu et al. (2021). XLS-R: Self-supervised Cross-lingual Speech Representation Learning at Scale. arXiv:2111.09296.
- Pascu et al. (2024). Calibrated deepfake detection in speech (cited in proposal).
- VaaniQ Capstone Proposal (2026). Cross-lingual compression-robust detection in Indian languages.
- Sharma, D. V., Ekbote, V., and Gupta, A. (2025). IndicSynth. ACL 2025, acl-long.1070.
- Javed, T. et al. (2022). IndicSUPERB (including Kathbath). arXiv:2208.11761.
- Mozilla Common Voice v17 (Hindi, Marathi).